Project: Worked examples
Title: Thick-walled cylinder stresses (Lamé)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Radial, hoop and axial stresses at the bore and the outer surface of a thick-walled cylinder under internal and external pressure, the von Mises and Tresca equivalents, utilisation against an allowable, and the thin-wall estimate for comparison.
$A = (a^2 p_i - b^2 p_o)/(b^2 - a^2)$, $B = a^2 b^2 (p_i - p_o)/(b^2 - a^2)$; $\sigma_r = A - B/r^2$, $\sigma_\theta = A + B/r^2$; $\sigma_z = A$ for closed ends, 0 for open. von Mises from the three principal stresses; Tresca = max principal difference.
Ref: Roark's Formulas for Stress and Strain Ch. 13, Table 13.5 — Thick-walled cylinders
Ref: Shigley's Mechanical Engineering Design §3-14 — Stresses in pressurised cylinders

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
a	Inner radius	40	mm	drawing: DRG-CYL-02 rev B	0.1 … 5000 mm
b	Outer radius	60	mm	drawing: DRG-CYL-02 rev B	0.1 … 5000 mm
p_i	Internal pressure	0	MPa	assumption: Vented bore.	0 … 1000 MPa
p_o	External pressure	20	MPa	spec: SPEC-SUB-1 §2 external pressure	0 … 1000 MPa
closed	Closed ends (axial stress carried by the wall)	yes		drawing: DRG-CYL-02 rev B	any
sigma_allow	Allowable stress	100	MPa	spec: SPEC-SUB-1 §4	1 … 3000 MPa

3. Assumptions

A1. Long cylinder, elastic, isotropic; stresses vary with radius only. (Lamé solution; end effects neglected.)
A2. Closed ends carry the pressure end load as a uniform axial stress. ($\sigma_z = A$.)
A3. p_i (Internal pressure) is taken as 0 MPa. (Vented bore.)

4. Working

```
[7] A = (a^2 * p_i - b^2 * p_o) / (b^2 - a^2)                    Roark Table 13.5
= ((40 mm)^2 × 0 MPa - (60 mm)^2 × 20 MPa) / ((60 mm)^2 - (40 mm)^2)
= -36 MPa
```
Lamé constant A

```
[8] B = a^2 * b^2 * (p_i - p_o) / (b^2 - a^2)                    Roark Table 13.5
= (40 mm)^2 × (60 mm)^2 × (0 MPa - 20 MPa) / ((60 mm)^2 - (40 mm)^2)
= -57600 MPa·mm²
```
Lamé constant B

```
[9] sigma_r_a = A - B / a^2
= (-36 MPa) - (-57600 MPa·mm²) / (40 mm)^2
= 0 MPa
```
Radial stress at the bore

```
[10] sigma_t_a = A + B / a^2
= -36 MPa + (-57600 MPa·mm²) / (40 mm)^2
= -72 MPa
```
Hoop stress at the bore
```
[11] sigma_r_b = A - B / b^2
= (-36 MPa) - (-57600 MPa·mm²) / (60 mm)^2
= -20 MPa
```
Radial stress at the outer surface
```
[12] sigma_t_b = A + B / b^2
= -36 MPa + (-57600 MPa·mm²) / (60 mm)^2
= -52 MPa
```
Hoop stress at the outer surface
```
[13] sigma_z = A
= -36 MPa
= -36 MPa
```
Axial stress (closed ends)
```
[14] sigma_vm_a = sqrt(((sigma_t_a - sigma_r_a)^2 + (sigma_r_a - sigma_z)^2 + (sigma_z - sigm
2)
= sqrt((((-72 MPa) - 0 MPa)^2 + (0 MPa - (-36 MPa))^2 + ((-36 MPa) - (-72 MPa))^2) / 2)
= 62.354 MPa
```
von Mises stress at the bore
```
[15] sigma_vm_b = sqrt(((sigma_t_b - sigma_r_b)^2 + (sigma_r_b - sigma_z)^2 + (sigma_z - sigm
2)
= sqrt((((-52 MPa) - (-20 MPa))^2 + ((-20 MPa) - (-36 MPa))^2 + ((-36 MPa) - (-52 MPa))^2) /
= 27.713 MPa
```
von Mises stress at the outer surface
```
[16] sigma_tr_a = max(sigma_t_a, sigma_r_a, sigma_z) - min(sigma_t_a, sigma_r_a, sigma_z)
= max(-72 MPa, 0 MPa, -36 MPa) - min(-72 MPa, 0 MPa, -36 MPa)
= 72 MPa
```
Tresca stress at the bore
```
[17] sigma_tr_b = max(sigma_t_b, sigma_r_b, sigma_z) - min(sigma_t_b, sigma_r_b, sigma_z)
= max(-52 MPa, -20 MPa, -36 MPa) - min(-52 MPa, -20 MPa, -36 MPa)
= 32 MPa
```
Tresca stress at the outer surface
```
[18] sigma_vm_max = max(sigma_vm_a, sigma_vm_b)
= max(62.354 MPa, 27.713 MPa)
= 62.354 MPa
```
Maximum von Mises stress
```
[19] u = sigma_vm_max / sigma_allow
= 62.354 MPa / 100 MPa
= 0.62354
```
Utilisation
```
[20] sigma_thin = (p_i - p_o) * (a + b) / 2 / (b - a)
= (0 MPa - 20 MPa) × (40 mm + 60 mm) / 2 / (60 mm - 40 mm)
= -50 MPa
```
Thin-wall hoop estimate at the mean radius
```
[21] k_wall = b / a
= 60 mm / 40 mm
= 1.5
```
Wall ratio b/a

5. Results

SYMBOL	VALUE	DESCRIPTION
sigma_r_a	0 MPa	Radial stress, bore

SYMBOL	VALUE	DESCRIPTION
sigma_t_a	-72 MPa	Hoop stress, bore
sigma_r_b	-20 MPa	Radial stress, outer
sigma_t_b	-52 MPa	Hoop stress, outer
sigma_z	-36 MPa	Axial stress
sigma_vm_a	62.354 MPa	von Mises, bore
sigma_vm_b	27.713 MPa	von Mises, outer
sigma_tr_a	72 MPa	Tresca, bore
sigma_tr_b	32 MPa	Tresca, outer
sigma_vm_max	62.354 MPa	Maximum von Mises
u	0.62354	Utilisation
sigma_thin	-50 MPa	Thin-wall hoop estimate
k_wall	1.5	Wall ratio

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	von Mises against allowable	0.62354	1	0.624	

7. Applicability

No warnings.

8. Verification

Template mech.thick-cylinder@2.0.0 Content hash c5a55d16b1831ed8 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0